Localization

State of art methods

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Introduction

What is Localization? I

Definition

- Localization is the fundamental process of accurately determining the **pose** (position and orientation) of an agent, object, or source within a given coordinate frame or map.[Pla+23]
- It answers the question: Where am I, and which way am I facing?

Analogy: The GPS Problem

- When you check your GPS to match your current sensor readings to a map's coordinates, you are solving the **localization** problem.[Pla+23]

High-level categories of SOTA methods

Categories (short) I

- **Global retrieval / image-based place recognition** — fast coarse search (NetVLAD-style descriptors).
- **Hierarchical (coarse-to-fine) localization** — retrieve candidates, then local matching & PnP / pose refinement (HF-Net, hloc pipelines).
- **Dense matching / synthesis verification** — dense descriptors and synthesized views for verification (InLoc).
- **Feature matching networks** — learned detectors + matchers (SuperPoint, SuperGlue, LoFTR, LightGlue).
- **SLAM / VIO hybrids** — combine continuous tracking (VO/VIO) with absolute localization from retrieval or map matching.

Representative methods (details)

NetVLAD (global retrieval) I

- Idea: train a CNN end-to-end with a VLAD-like pooling layer for place recognition at scale.
- Use as first stage: identify candidate reference images for a query (efficient indexing).
- Citation: [Ara+16].

HF-Net / hierarchical localization I

- Monolithic network predicting global descriptor + dense local descriptors + keypoint scores.
- Pipeline: global retrieval → local matching (learned features) → geometric verification / PnP.
- Suited to large-scale 6-DoF localization; trades off speed and robustness by narrowing search space early.
- Citation: [Sar+19].

SuperPoint + SuperGlue (matching backend) I

- SuperPoint: learned interest points + descriptors; SuperGlue: graph-neural matching + optimal transport-style solver to produce robust matches.
- Use: substitute for classical detector+descriptor+ratio-test in SfM/SLAM pipelines, or inside hierarchical systems.
- Citation: [Sar+20].

InLoc (dense matching + view synthesis) I

- Dense pixel-level matching and virtual view synthesis to verify candidate poses in indoor/textureless scenes.
- Strong for indoor localization where sparse keypoints fail.
- Citation: [TOS+18].

Practical implementation & integration notes

Pipeline recipe (practical) I

1. Build reference map: capture images, compute global descriptors (NetVLAD or global descriptors) and a 3D model (SfM).
2. At runtime: compute query global descriptor → retrieve top-K reference images.
3. Compute dense/local matches (SuperGlue/LoFTR/SuperPoint) between query and selected references.
4. Geometric verification: PnP with RANSAC → optimization (Bundle Adjustment) → pose.
5. Optionally fuse with continuous tracking (VO/VIO) for smoothness and drift correction.

Embedded considerations (Jetson/Pi) I

- Use lightweight backbones (MobileNet, TinyViT) and convert to TensorRT (FP16/INT8) for Jetson inference.
- Run global retrieval at reduced frequency, and run heavy matching only when necessary.
- Keep descriptor databases on fast storage; use approximate nearest neighbor libraries (FAISS) tuned for memory footprint.

SLAM and Active SLAM

Simultaneous Localization and Mapping (SLAM) I

The SLAM Problem

- The agent must **localize** itself in an unknown environment while simultaneously **building a map**.
- Key paradigms include **Filter-based** (e.g., EKF-SLAM) and **Optimization-based** (e.g., Graph-SLAM).

Active SLAM

- An advanced paradigm where the agent **actively decides its path** to reduce uncertainty (information-theoretic planning).
- This area is a key frontier in robotics research [Pla+23].

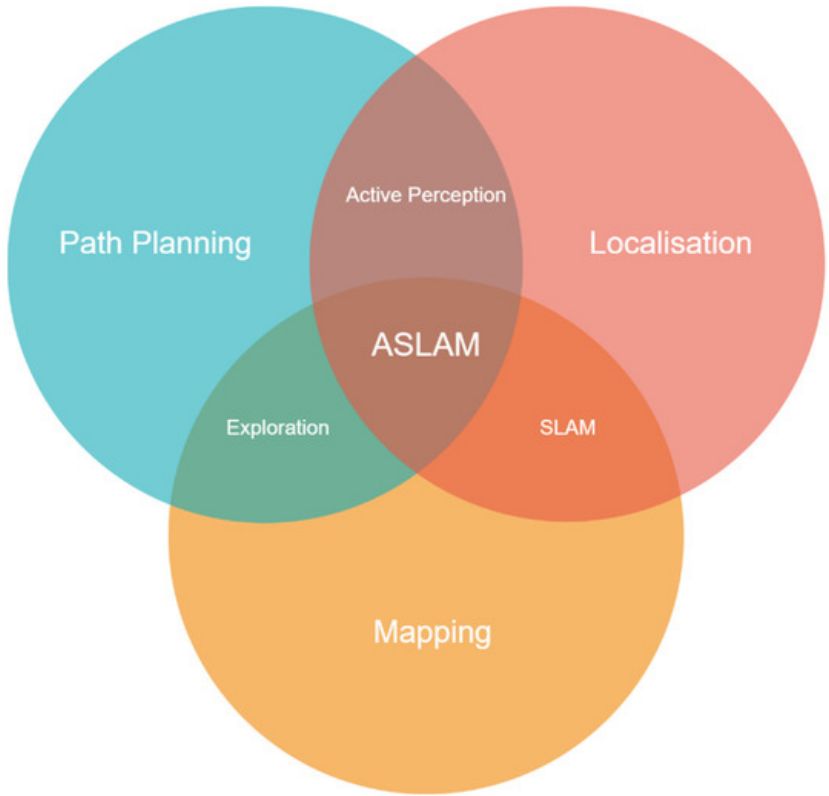


Abbildung: Active SLAM Venn Diagram [Mih+22]

Simultaneous Localization and Mapping I

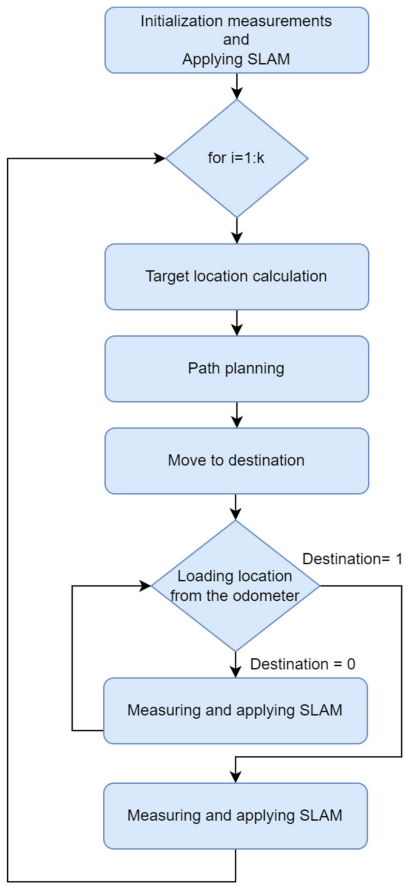


Abbildung: SLAM Flowchart [Mih+22]

Multi-Agent and Distributed Localization

Distributed Localization for Multi-Robot Systems I

Motivation

- Necessary for swarms, collaborative tasks, and robust large-area coverage.
- Emphasizes resilience and scalability by avoiding centralized processing.

Distributed Relative Localization

- Robots use **local information** (e.g., inter-robot distance/bearing) to estimate their own position and the positions of their neighbors.
- Algorithms must manage **communication latency** and **non-global consensus** for network coordination [Wan+23].

Deep Learning in Localization

Multi-Time Pressure and Eigenvector Feature(MTPEF) I

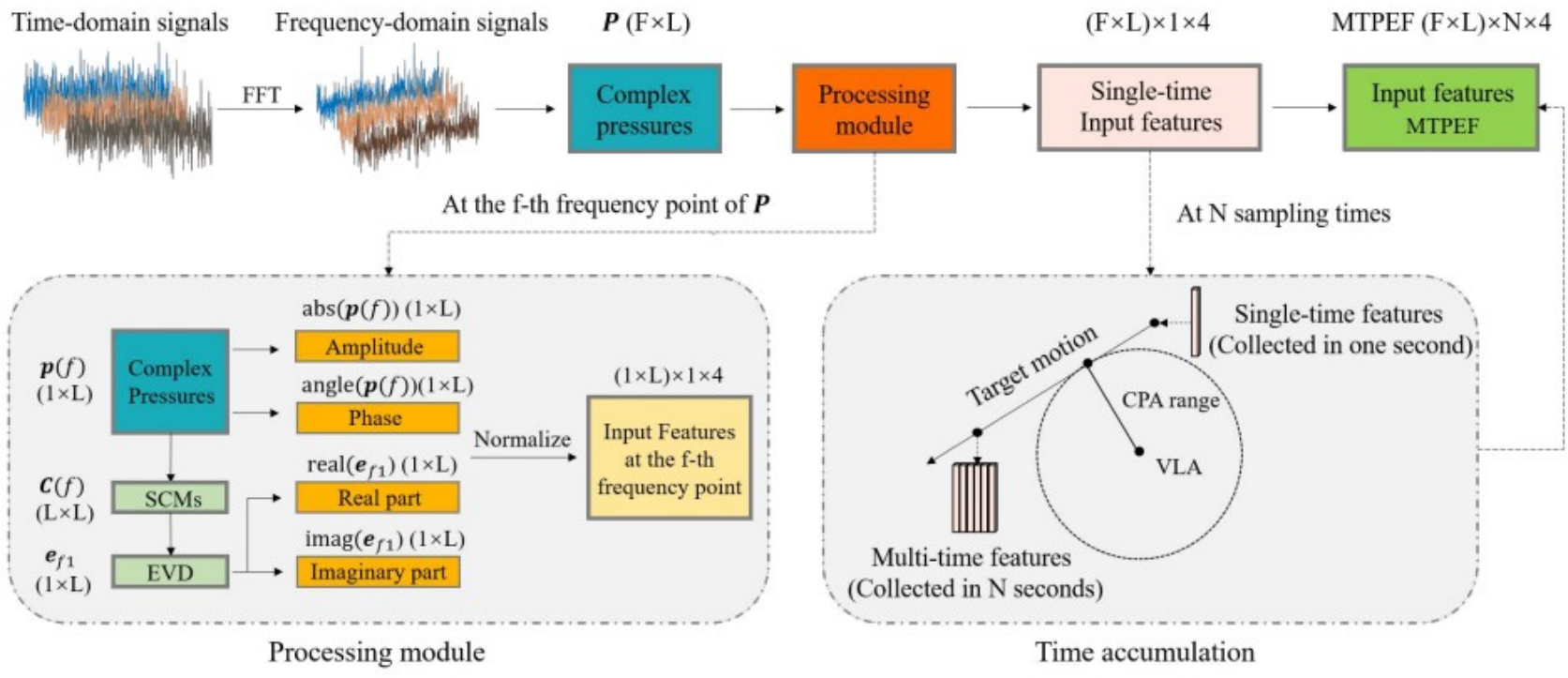


Abbildung: Multi-Time Pressure and Eigenvector Feature [SFT24]

Deep Learning for Localization and Mapping (DL-LM) I

The Data-Driven Approach

- Uses neural networks to **learn the localization function** directly from raw sensor data, offering robustness over hand-engineered models.

Applications

- **Visual SLAM/Odometry:** Learning pose regression or feature extraction from camera data [Che+23].
- **Acoustic Source Localization:** Using DNNs for direction of arrival (DoA) estimation in noisy environments [Gru+22] and via improved input features [SFT24].
- **Microscopy Localization:** Applying lightweight DL for precise localization of single molecules in high-resolution bioimaging [Fei+25].

Network Structure: Inception and Residual I

The IRSNet Architecture

- **IRSNet** (Inception and Residual in-series Network) is the network structure designed specifically for the deep learning acoustic source localization method.
- **Inception Modules:** These are incorporated to effectively capture feature information at **multiple scales** from the MTPEF input features.
- **Residual Modules:** These are used to facilitate the training of a **deep network** by creating shortcut connections (skip connections) that help prevent the problem of gradient degradation and diminishing returns as the network depth increases.
- **Goal:** This tailored combination significantly **boosts localization accuracy** and robustness, especially for underwater acoustic characteristics.[SFT24]

IRSNet Architecture I

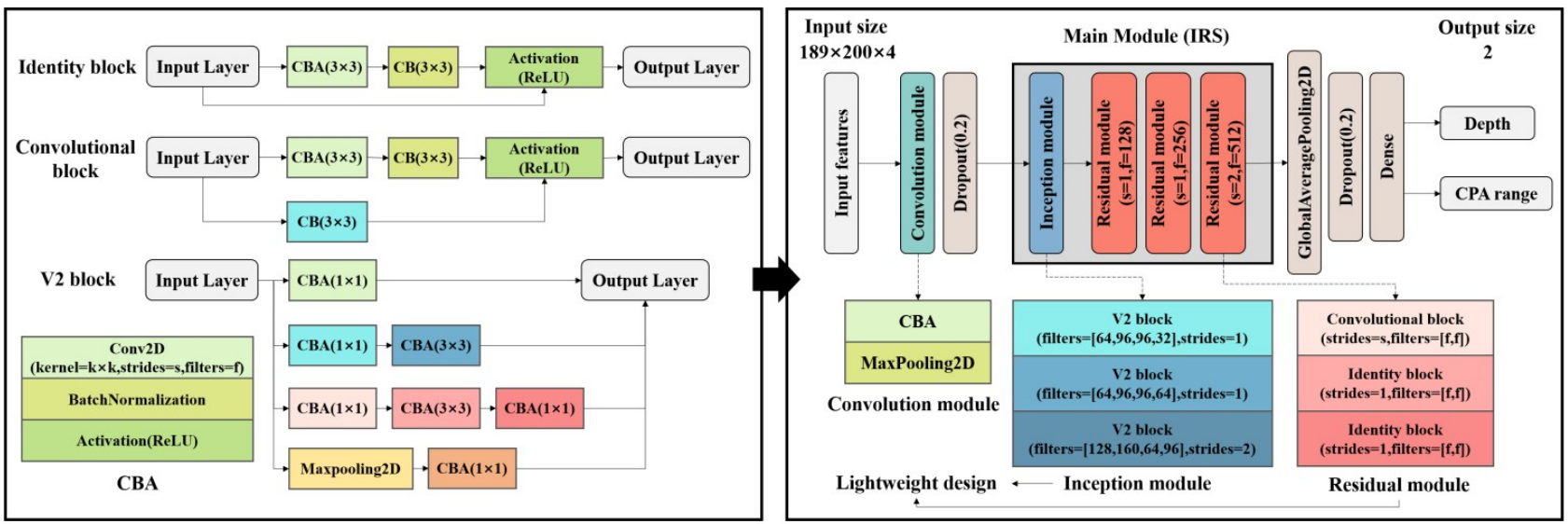


Abbildung: Inception and Residual in-series Network [SFT24]

Vielen Dank

für Ihre Aufmerksamkeit

Quellen

Quellen I

Nachfolgend werden die Quellen der Bilder angegeben, die für diese Präsentation in ihrer ursprünglichen Form oder modifiziert verwendet worden sind.

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